

Information Field Theory

Final Synthesis and Unification

IFT-Synthesis v1.0

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The final synthesis of the Information Field Theory (IFT) project is presented, unifying the 14 developed modules into a single coherent framework. IFT is a field theory based on a positive scalar field $\rho(x) > 0$ representing relational information density. From this single primitive field, the following are derived: the mathematical core (IFT-Core) with 9 fundamental parameters and 13 master equations; consciousness (IFT-C) with the functional Φ_{IFT} and the dual criterion $\Phi_{\text{IFT}} > \Phi_c \wedge \tau\Xi > K$; quantum biology (IFT-Bio) with the spectral density $J_{\text{IFT}}(\omega)$ and the scaling hypothesis $\tau_\phi\omega_c \approx K_Q$; geometry and gravity (IFT-Grav) with the informational metric $\tilde{g}_{\mu\nu}[\rho]$ and the gravitational-wave correction $\kappa_1 \sim 10^{-15}$; cosmology (IFT-Cosmo) with $w(z) = -1 + \varepsilon z/(1+z)$ and $\varepsilon \approx 0.12 - 0.18$; matter and mass (IFT-Mass) with the entropic mechanism $M = m_P e^{-S/2}$; Yang-Mills (IFT-YM) with the hierarchical cluster expansion toward the mass gap; and extensions to thermodynamics, economics, neuroscience, quantum information, philosophy, evolution, and climatology. A master table of falsifiable predictions and the status of each module are presented.

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Chapter 1

Introduction: The IFT Project

1.1 What is IFT?

Information Field Theory (IFT) is a field theory based on a single positive scalar field:

$$\rho : M \rightarrow \mathbb{R}_{>0}, \quad \rho \in C^{2,\alpha}(M) \cap L^1(M). \quad (1.1)$$

The fundamental hypothesis is that all observable physics can be organized around this relational information field.

1.2 Project Architecture

IFT is structured at three levels:

1. **Level 1 – Core:** IFT-Core (Document 1). Objects, axioms, action, equations of motion, informational geometry, quantization.
2. **Level 2 – Physical Extensions:** IFT-Consciousness (2), IFT-Bio (3), IFT-Grav (4), IFT-Cosmo (5), IFT-Mass (6), IFT-YM (7).
3. **Level 3 – Interdisciplinary Extensions:** IFT-Thermo (8), IFT-Econ (9), IFT-Neuro (10), IFT-QInfo (11), IFT-Phil (12), IFT-Evo (13), IFT-Climate (14).

Chapter 2

Summary of Each Module

2.1 IFT-Core (Document 1)

Thesis: There exists a field $\rho > 0$ from which all physical structures emerge.

Key equations: $Z_\rho \square u + V'(u) = 0$, $I[\rho] = 4 \int (\nabla u)^2 d\mu$, $\tilde{g}_{\mu\nu} = H_{\mu\nu} - \lambda_F I_{\mu\nu}$.

Theorems: Global existence, vacuum stability, mass bound $m_n \geq C \frac{\hbar}{c} \sqrt{\langle \Xi^2 \rangle_n}$, emergent Lorentzian signature, topological soliton classification.

Parameters: 9 ($G, \Lambda_0, Z_\rho, \lambda, v, \eta, \kappa_\rho, \lambda_F, \sigma$).

2.2 IFT-Consciousness (Document 2)

Thesis: Consciousness is informational self-presence with criterion $\Phi_{\text{IFT}} > \Phi_c \wedge \tau \Xi > K$.

Key equation: $\Phi_{\text{IFT}} = \int w(x) [\alpha_I \rho \ln(\rho/\rho_{\text{base}})/\rho_* + \alpha_F \ell_F^2 \text{Tr}(\kappa_F) + \alpha_C I_{\text{causal}}] dV$.

Validation: Consistent with Cogitate (2025) and IIT/GNWT tests.

2.3 IFT-Bio (Document 3)

Thesis: Functional quantum coherence arises from bath organization, not isolation.

Key equation: $J_{\text{IFT}}(\omega) = J_0(\omega/\omega_c)^s e^{-\omega/\omega_c}$, $s = 1 + D_f$, $\tau_\phi \omega_c \approx K_Q$.

Validation: FMO coherence at room temperature (Cao 2024), vibronic enzymatic tunneling (Klinman 2025), MagR (Xie 2025), quantum olfaction (Franco 2025).

2.4 IFT-Grav (Document 4)

Thesis: Physical geometry is coupled to the informational geometry of ρ .

Key equation: $h(f) = h_{\text{GR}}(f)[1 + \kappa_1(f/f_{\text{ref}})]$, $\kappa_1 \sim 10^{-15}$.

Validation: Consistent with LIGO bounds ($\kappa_1 < 4.9 \times 10^{-14}$). Testable with ET/CE.

2.5 IFT-Cosmo (Document 5)

Thesis: The field $u(t)$ provides dynamical dark energy with $w(z) = -1 + \varepsilon z/(1+z)$.

Key equation: $Z_\rho(\ddot{u} + 3H\dot{u}) + V'(u) = 0$, $\varepsilon \approx 0.12 - 0.18$.

Validation: Consistent with DESI DR2 (2024, 2.5σ). Testable with Euclid/CMB-S4.

2.6 IFT-Mass (Document 6)

Thesis: Mass is the spectral gap of stable informational configurations.

Key equation: $M = m_P e^{-S/2}$, $m_\psi = y_\rho u_0$, $m_W = g_2 v/2$.

Status: Correct functional form of the hierarchy. Specific values not derived.

2.7 IFT-YM (Document 7)

Thesis: The Yang-Mills mass gap can be attacked with a positive informational regulator.

Key equation: $|\rho(\gamma)| \leq e^{-A_0 m_u \cdot \text{diam}(\gamma) - B_0 |\gamma|}$.

Status: Theorems proved (existence, gauge, OS2, gap for large m_u). Complete conditional chain. Mass gap NOT proved. Clay problem open.

Chapter 3

Interdisciplinary Extensions (Documents 8–14)

- **IFT-Thermo:** H-theorem, $T_{\text{info}} = (\partial S / \partial E)^{-1}$, connection with Hawking.
- **IFT-Econ:** Prices $\propto \Xi$, crashes as $\tau \Xi < K_{\text{market}}$.
- **IFT-Neuro:** Experimental protocols competitive with IIT/GNWT.
- **IFT-QInfo:** $S_{EE} \leq C \int_{\partial A} \Xi dA$, complexity $\propto \int \Xi dx$.
- **IFT-Phil:** Informational structural realism, ontology of ρ .
- **IFT-Evo:** Fitness $= -D_{\text{KL}}(\rho || \rho^*)$, speciation as a phase transition.
- **IFT-Climate:** Tipping points as $\tau \Xi < K_{\text{climate}}$.

Chapter 4

Master Table of Falsifiable Predictions

Domain	IFT Prediction	Observable	Experiment	Timeframe
Gravitational	$h(f) = h_{\text{GR}}(f)[1 + \kappa_1 f/f_{\text{ref}}]$	$\kappa_1 \sim 10^{-15}$	LIGO/ET/CE	2025–2035
Cosmology	$w(z) = -1 + \varepsilon z/(1+z)$	$\varepsilon \approx 0.15$	DESI/Euclid/CMB-S4	2025–2035
Structure	$S_8 \approx 0.80$	S_8	Euclid/LSST	2027–2035
Particles	Dilaton with broad m_u	$g_{u\gamma\gamma} \sim 10^{-12} \text{ GeV}^{-1}$	ALPS II/IAXO	2025–2035
Consciousness	$\Phi_{\text{IFT}} > \Phi_c \wedge \tau\Xi > K$	Φ_{IFT} from EEG/fMRI	Multisite study	2025–2035
Anesthesia	$\Phi_{\text{IFT}} \downarrow$ with propofol	$\Phi_{\text{IFT}}(t)$	Anesthesia protocol	2025–2035
Quantum Biology	$\tau_\phi \omega_c \approx K_Q$	τ_ϕ in FMO	2D spectroscopy	2025–2035
Evolution	D_f increases under selection	D_f of proteins	Experimental evolution	2026–2035
Climate	$\tau\Xi$ predicts tipping points	$\tau\Xi$ AMOC	Oceanographic data	2025–2035
Economics	Crash when $\tau\Xi < K_{\text{market}}$	$\tau\Xi$ stock indices	Financial data	2025–2035
Quantum Info	$S_{EE} \leq C \int_{\partial A} \Xi dA$	S_{EE} in qubits	Tomography	2025–2035

Table 4.1: Master table of falsifiable IFT predictions.

Chapter 5

Summary of Proved Theorems

Theorem	Statement	Module
Lattice existence	$Z_a < \infty$ in finite volume	Core, YM
Gauge invariance	$S_a[hUh^{-1}, u] = S_a[U, u]$	YM
Pure YM limit	$S_a \rightarrow S_W$ when $u \rightarrow v$	YM
Reflection positivity	$\langle \Theta F \cdot F \rangle \geq 0$	YM, Core
Lorentzian signature	$\tilde{g}_{\mu\nu}$ has $(-, +, +, +)$ locally	Core, Grav
Fisher identity	$\int (\nabla \rho)^2 / \rho = 4 \int (\nabla u)^2$	Core
Global existence	Global solution for $Z_\rho \square u + V'(u) = 0$	Core
Vacuum stability	$u = v$ is stable under small perturbations	Core
Mass bound	$m_n \geq C \frac{\hbar}{c} \sqrt{\langle \Xi^2 \rangle_n}$	Core
Vacuum uniqueness	$\hat{H}$ has a unique ground state	Core
Scalar gap	$\text{Spec}(\hat{H}_\rho) = \{0\} \cup [m_u c^2, \infty)$	Core
Generalized Derrick	Solitons exist for $\kappa_\rho > 0$	Core, Mass
Q-ball existence	Charge-fixed minimizer exists	Core
BPS bound	$M_n \geq \frac{4\pi v}{g} n $	Core
Activity bound	$ \rho(\gamma) \leq e^{-A_0 m_u \text{diam} - B_0 \gamma }$	YM
Strong clustering	Exponential decay for large m_u	YM
H-theorem	$dS/dt \geq 0$ for gradient flow	Thermo

Chapter 6

Open Conjectures

1. **Mass Gap (Clay):** Prove $\liminf_{a \rightarrow 0} m_{\text{lat}}(a)/a > 0$ for pure Yang-Mills.
2. **Uniform continuum limit:** Non-perturbative RG control of the cluster expansion.
3. **Stable decoupling:** Gap preservation when $u \rightarrow v$.
4. **Complete OS reconstruction:** Emergent Lorentz invariance.
5. **Derivation of masses:** Compute y_f, λ_H, v from first principles.
6. **Calibration of Φ_c, K :** Experimental determination of consciousness thresholds.
7. **Cosmological constant:** Derive $\Lambda_{\text{eff}} \approx 10^{-52} \text{ m}^{-2}$ without fine-tuning.
8. **Gauge group selection:** Uniquely derive $SU(3) \times SU(2) \times U(1)$.

Chapter 7

Future Research Program

7.1 Short Term (2025–2027)

- Publish IFT-Core, IFT-C, IFT-Bio, IFT-Grav, IFT-Cosmo in theoretical physics journals.
- Conduct a consciousness study with $N = 100$ to calibrate Φ_c and K .
- Verify the prediction $\tau_\phi \omega_c \approx K_Q$ in FMO with mutants.

7.2 Medium Term (2027–2030)

- Analyze Euclid and CMB-S4 data to test $w(z) \neq -1$.
- Search for κ_1 in LIGO O4/O5 data.
- Develop lattice simulations of IFT-YM to verify clustering.

7.3 Long Term (2030–2040)

- Detection of the IFT dilaton at ALPS II/IAXO or colliders.
- Consciousness tests on advanced neuromorphic systems.
- Rigorous proof of the mass gap (if the conjectures are verified).

Chapter 8

Universal Master Formula of IFT

$$\boxed{\rho(x) > 0, \quad u = \sqrt{\rho}, \quad \varphi = \ln \rho, \quad \Xi_\mu = \nabla_\mu \ln \rho}. \quad (8.1)$$

$$\boxed{S_{\text{IFT}} = \int d^4x \sqrt{-g} \left[\frac{R - 2\Lambda_0}{16\pi G} + \frac{Z_\rho}{2} (\nabla u)^2 - V(u) - \frac{\kappa_\rho}{2} (\Box u)^2 + \frac{1}{2\sigma^2} \|g - \tilde{g}[\rho]\|^2 \right]}. \quad (8.2)$$

$$\boxed{m_n \geq C \frac{\hbar}{c} \sqrt{\langle \Xi^2 \rangle_n}, \quad M = m_P e^{-S/2}, \quad \Phi_{\text{IFT}} > \Phi_c \wedge \tau \Xi > K}. \quad (8.3)$$

Chapter 9

Final Declaration

Information Field Theory is a unifying framework based on the premise that relational information, encoded in the field $\rho(x) > 0$, is the fundamental ontological category. From this single field and 9 parameters, IFT provides:

- A rigorous mathematical core with proved theorems of existence, stability, and spectral bounds.
- Extensions to gravity, gauge theory, matter, cosmology, consciousness, and quantum biology.
- Falsifiable predictions in gravitational waves, dark energy, particle physics, and neuroscience.
- Interdisciplinary applications to thermodynamics, economics, evolution, and climatology.
- A concrete route toward the Yang-Mills mass gap (not solved, but structurally formulated).

IFT does not claim to be a closed “theory of everything.” It claims to be a legitimate scientific framework: it makes quantitative predictions, accepts refutation criteria, and clearly distinguishes between proved theorems, conditional hypotheses, and open conjectures.

IFT = $\rho(x) > 0$ as the foundation of physics and beyond

The theory will be strong if it predicts. It will be scientific if it accepts refutation.

Chapter 10

Complete List of IFT Documents

1. **IFT-Core v2.0**: Definitive Fundamental Core.
2. **IFT-Consciousness v2.0**: Operational dissolution of the hard problem.
3. **IFT-Quantum Biology v2.0**: Experimental validation 2024–2026.
4. **IFT-Geometry & Gravity v2.0**: Informational geometry and gravity.
5. **IFT-Cosmology v2.0**: Dynamical dark energy and DESI data.
6. **IFT-Matter & Mass v2.0**: Spectral origin of mass.
7. **IFT-Yang-Mills v2.0**: Program toward the mass gap.
8. **IFT-Thermodynamics v1.0**: Informational H-theorem.
9. **IFT-Econ v1.0**: Markets as informational fields.
10. **IFT-Neuroscience v1.0**: Experimental protocols for Φ_{IFT} .
11. **IFT-QuantumInfo v1.0**: Entanglement and complexity.
12. **IFT-Philosophy v1.0**: Informational structural realism.
13. **IFT-Evolution v1.0**: Fitness as KL divergence.
14. **IFT-Climate v1.0**: Tipping points as phase transitions.
15. **IFT-Synthesis v1.0**: Final synthesis and unification (this document).